1. Create a new database named 'ADS' and switch to it. Then, create a collection called 'employees' and insert the following records.

```
test> show dbs
admin      40.00 KiB
company    72.00 KiB
companyDB  68.00 KiB
config     72.00 KiB
local      64.00 KiB
schoolDB   68.00 KiB
test> use ADS
switched to db ADS
ADS> db
ADS
ADS> db.employees.insertMany([
|    { name: "Ana Cruz", department: "IT", salary: 32000, age: 25 },
|    { name: "Mark Lee", department: "HR", salary: 28000, age: 23 },
|    { name: "John Reyes", department: "IT", salary: 35000, age: 28 },
|    { name: "Lisa Tan", department: "Finance", salary: 30000, age: 21 },
|    { name: "Paul Kim", department: "IT", salary: 40000, age: 35 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69cb48a585c4163d5d11402a'),
    '1': ObjectId('69cb48a585c4163d5d11402b'),
    '2': ObjectId('69cb48a585c4163d5d11402c'),
    '3': ObjectId('69cb48a585c4163d5d11402d'),
    '4': ObjectId('69cb48a585c4163d5d11402e')
  }
}
ADS> |
```

Task 1: Display all employees sorted by salary (descending).

```
ADS> db.employees.find().sort({ salary: -1 })
[
  {
    _id: ObjectId('69cb48a585c4163d5d11402e'),
    name: 'Paul Kim',
    department: 'IT',
    salary: 40000,
    age: 35
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402c'),
    name: 'John Reyes',
    department: 'IT',
    salary: 35000,
    age: 28
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402a'),
    name: 'Ana Cruz',
    department: 'IT',
    salary: 32000,
    age: 25
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402d'),
    name: 'Lisa Tan',
    department: 'Finance',
    salary: 30000,
    age: 21
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402b'),
    name: 'Mark Lee',
    department: 'HR',
    salary: 28000,
    age: 23
  }
]
ADS> |
```

2. Find employees with salary greater than 30,000 AND age less than 30.

```
ADS> db.employees.find({ $and: [ { salary: { $gt: 30000 } }, { age: { $lt: 30 } } ] })
[
  {
    _id: ObjectId('69cb48a585c4163d5d11402a'),
    name: 'Ana Cruz',
    department: 'IT',
    salary: 32000,
    age: 25
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402c'),
    name: 'John Reyes',
    department: 'IT',
    salary: 35000,
    age: 28
  }
]
```

3. Retrieve employees in IT OR Finance departments.

```
ADS> db.employees.find({ $or: [ { department: "IT" }, { department: "Finance" } ] })
[
  {
    _id: ObjectId('69cb48a585c4163d5d11402a'),
    name: 'Ana Cruz',
    department: 'IT',
    salary: 32000,
    age: 25
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402c'),
    name: 'John Reyes',
    department: 'IT',
    salary: 35000,
    age: 28
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402d'),
    name: 'Lisa Tan',
    department: 'Finance',
    salary: 30000,
    age: 21
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402e'),
    name: 'Paul Kim',
    department: 'IT',
    salary: 40000,
    age: 35
  }
]
ADS>
```

4. Find employees whose age is not equal to 25.

```
ADS> db.employees.find({ age: { $ne: 25 } })
[
  {
    _id: ObjectId('69cb48a585c4163d5d11402b'),
    name: 'Mark Lee',
    department: 'HR',
    salary: 28000,
    age: 23
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402c'),
    name: 'John Reyes',
    department: 'IT',
    salary: 35000,
    age: 28
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402d'),
    name: 'Lisa Tan',
    department: 'Finance',
    salary: 30000,
    age: 21
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402e'),
    name: 'Paul Kim',
    department: 'IT',
    salary: 40000,
    age: 35
  }
]
ADS>
```

5. Update one employee named "Lisa Tan" and set salary to 32000.

```
ADS> db.employees.updateOne({ name: "Lisa Tan" }, { $set: { salary: 32000 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
ADS> db.employees.find({ name: "Lisa Tan" })
[
  {
    _id: ObjectId('69cb48a585c4163d5d11402d'),
    name: 'Lisa Tan',
    department: 'Finance',
    salary: 32000,
    age: 21
  }
```

6. Update all employees with salary greater than 30,000 and add status: "high-paid".

```
]
ADS> db.employees.updateMany({ salary: { $gt: 30000 } }, { $set: { status: "high-paid" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
ADS> db.employees.find({ status: "high-paid" })
[
  {
    _id: ObjectId('69cb48a585c4163d5d11402a'),
    name: 'Ana Cruz',
    department: 'IT',
    salary: 32000,
    age: 25,
    status: 'high-paid'
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402c'),
    name: 'John Reyes',
    department: 'IT',
    salary: 35000,
    age: 28,
    status: 'high-paid'
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402d'),
    name: 'Lisa Tan',
    department: 'Finance',
    salary: 32000,
    age: 21,
    status: 'high-paid'
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402e'),
    name: 'Paul Kim',
    department: 'IT',
    salary: 40000,
    age: 35,
    status: 'high-paid'
  }
]
ADS> |
```

7. Delete employees with age less than 23.

```
]
ADS> db.employees.find({ age: { $lt: 23 } })
[
  {
    _id: ObjectId('69cb48a585c4163d5d11402d'),
    name: 'Lisa Tan',
    department: 'Finance',
    salary: 32000,
    age: 21,
    status: 'high-paid'
  }
]
ADS> db.employees.deleteMany({ age: { $lt: 23 } })
{ acknowledged: true, deletedCount: 1 }
ADS> db.employees.find()
[
  {
    _id: ObjectId('69cb48a585c4163d5d11402a'),
    name: 'Ana Cruz',
    department: 'IT',
    salary: 32000,
    age: 25,
    status: 'high-paid'
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402b'),
    name: 'Mark Lee',
    department: 'HR',
    salary: 28000,
    age: 23
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402c'),
    name: 'John Reyes',
    department: 'IT',
    salary: 35000,
    age: 28,
    status: 'high-paid'
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402e'),
    name: 'Paul Kim',
    department: 'IT',
    salary: 40000,
    age: 35,
    status: 'high-paid'
  }
]
ADS> |
```

8. Count employees in the IT department.

```
ADS> db.employees.countDocuments({ department: "IT" })
3
```

```
ADS> db.employees.find({ department: "IT" })
[
  {
    _id: ObjectId('69cb48a585c4163d5d11402a'),
    name: 'Ana Cruz',
    department: 'IT',
    salary: 32000,
    age: 25,
    status: 'high-paid'
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402c'),
    name: 'John Reyes',
    department: 'IT',
    salary: 35000,
    age: 28,
    status: 'high-paid'
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402e'),
    name: 'Paul Kim',
    department: 'IT',
    salary: 40000,
    age: 35,
    status: 'high-paid'
```

9. Limit the result to top 3 highest-paid employees.

```
ADS> db.employees.find().sort({ salary: -1 }).limit(3)
[
  {
    _id: ObjectId('69cb48a585c4163d5d11402e'),
    name: 'Paul Kim',
    department: 'IT',
    salary: 40000,
    age: 35,
    status: 'high-paid'
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402c'),
    name: 'John Reyes',
    department: 'IT',
    salary: 35000,
    age: 28,
    status: 'high-paid'
  },
  {
    _id: ObjectId('69cb48a585c4163d5d11402a'),
    name: 'Ana Cruz',
    department: 'IT',
    salary: 32000,
    age: 25,
    status: 'high-paid'
```

10. Create an index on the department field and display all indexes.

```
ADS> db.employees.createIndex({ department: 1 })
department_1
ADS> db.employees.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { department: 1 }, name: 'department_1' }
]
ADS>
```

In the same database, create a new collection named 'students' and insert the records below.

```
ADS> db.students.insertMany([
|   { name: "Alice", age: 19, grade: 85 },
|   { name: "Bob", age: 17, grade: 90 },
|   { name: "Charlie", age: 22, grade: 70 },
|   { name: "Diana", age: 20, grade: 95 },
|   { name: "Eve", age: 18, grade: 60 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69cb4d8d85c4163d5d11402f'),
    '1': ObjectId('69cb4d8d85c4163d5d114030'),
    '2': ObjectId('69cb4d8d85c4163d5d114031'),
    '3': ObjectId('69cb4d8d85c4163d5d114032'),
    '4': ObjectId('69cb4d8d85c4163d5d114033')
  }
}
ADS>
```

Tasks: 1. Display all students sorted by grade (ascending).

```
}
ADS> db.students.find().sort({ grade: 1 })
[
  {
    _id: ObjectId('69cb4d8d85c4163d5d114033'),
    name: 'Eve',
    age: 18,
    grade: 60
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114031'),
    name: 'Charlie',
    age: 22,
    grade: 70
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d11402f'),
    name: 'Alice',
    age: 19,
    grade: 85
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114030'),
    name: 'Bob',
    age: 17,
    grade: 90
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114032'),
    name: 'Diana',
    age: 20,
    grade: 95
  }
]
ADS>
```

2. Find students with grade greater than or equal to 85.

```
ADS> db.students.find({ grade: { $gte: 85 } })
[
  {
    _id: ObjectId('69cb4d8d85c4163d5d11402f'),
    name: 'Alice',
    age: 19,
    grade: 85
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114030'),
    name: 'Bob',
    age: 17,
    grade: 90
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114032'),
    name: 'Diana',
    age: 20,
    grade: 95
  }
]
```

3. Retrieve students with age less than 20 OR grade below 75.

```
ADS> db.students.find({ $or: [ { age: { $lt: 20 } }, { grade: { $lt: 75 } } ] })
[
  {
    _id: ObjectId('69cb4d8d85c4163d5d11402f'),
    name: 'Alice',
    age: 19,
    grade: 85
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114030'),
    name: 'Bob',
    age: 17,
    grade: 90
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114031'),
    name: 'Charlie',
    age: 22,
    grade: 70
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114033'),
    name: 'Eve',
    age: 18,
    grade: 60
  }
]
ADS> |
```

4. Find students whose grade is not equal to 90.

```
ADS> db.students.find({ grade: { $ne: 90 } })
[
  {
    _id: ObjectId('69cb4d8d85c4163d5d11402f'),
    name: 'Alice',
    age: 19,
    grade: 85
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114031'),
    name: 'Charlie',
    age: 22,
    grade: 70
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114032'),
    name: 'Diana',
    age: 20,
    grade: 95
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114033'),
    name: 'Eve',
    age: 18,
    grade: 60
  }
```

5. Update student "Eve" and set grade to 75.

```
ADS> db.students.findOne({ name: "Eve" })
{
  _id: ObjectId('69cb4d8d85c4163d5d114033'),
  name: 'Eve',
  age: 18,
  grade: 60
}
ADS> db.students.updateOne({ name: "Eve" }, { $set: { grade: 75 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
ADS> db.students.findOne({ name: "Eve" })
{
  _id: ObjectId('69cb4d8d85c4163d5d114033'),
  name: 'Eve',
  age: 18,
  grade: 75
}
ADS> |
```

6. Update all students aged 18 and above, add status: "eligible".

```
}
ADS> db.students.find({ age: { $gte: 18 } })
[
  {
    _id: ObjectId('69cb4d8d85c4163d5d11402f'),
    name: 'Alice',
    age: 19,
    grade: 85
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114031'),
    name: 'Charlie',
    age: 22,
    grade: 70
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114032'),
    name: 'Diana',
    age: 20,
    grade: 95
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114033'),
    name: 'Eve',
    age: 18,
    grade: 75
  }
]
ADS> db.students.updateMany({ age: { $gte: 18 } }, { $set: { status: "eligible" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
ADS> db.students.find({ age: { $gte: 18 } })
[
  {
    _id: ObjectId('69cb4d8d85c4163d5d11402f'),
    name: 'Alice',
    age: 19,
    grade: 85,
    status: 'eligible'
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114031'),
    name: 'Charlie',
    age: 22,
    grade: 70,
    status: 'eligible'
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114032'),
    name: 'Diana',
    age: 20,
    grade: 95,
    status: 'eligible'
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114033'),
    name: 'Eve',
    age: 18,
    grade: 75,
    status: 'eligible'
  }
```

7. Delete students with grade less than 70.

```
]
ADS> db.students.find({ grade: { $lt: 70 } })

ADS> db.students.deleteMany({ grade: { $lt: 70 } })
{ acknowledged: true, deletedCount: 0 }
ADS> db.students.find()
[
  {
    _id: ObjectId('69cb4d8d85c4163d5d11402f'),
    name: 'Alice',
    age: 19,
    grade: 85,
    status: 'eligible'
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114030'),
    name: 'Bob',
    age: 17,
    grade: 90
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114031'),
    name: 'Charlie',
    age: 22,
    grade: 70,
    status: 'eligible'
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114032'),
    name: 'Diana',
    age: 20,
    grade: 95,
    status: 'eligible'
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114033'),
    name: 'Eve',
    age: 18,
    grade: 75,
    status: 'eligible'
  }
]
ADS>
```

since we updated student eve her grade become 60 to 75, therefore no student have with grade less than 70.

8. Count how many students have grade above 80

```
ADS> db.students.countDocuments({ grade: { $gt: 80 } })
3
ADS> db.students.find({ grade: { $gt: 80 } })
[
  {
    _id: ObjectId('69cb4d8d85c4163d5d11402f'),
    name: 'Alice',
    age: 19,
    grade: 85,
    status: 'eligible'
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114030'),
    name: 'Bob',
    age: 17,
    grade: 90
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114032'),
    name: 'Diana',
    age: 20,
    grade: 95,
    status: 'eligible'
  }
```

9. Display only 2 students with highest grades.

```
ADS> db.students.find().sort({ grade: -1 }).limit(2)
[
  {
    _id: ObjectId('69cb4d8d85c4163d5d114032'),
    name: 'Diana',
    age: 20,
    grade: 95,
    status: 'eligible'
  },
  {
    _id: ObjectId('69cb4d8d85c4163d5d114030'),
    name: 'Bob',
    age: 17,
    grade: 90
  }
]
ADS>
```

10. Create an index on the grade field and show all indexes.

```
ADS> db.students.createIndex({ grade: 1 })
grade_1
ADS> db.students.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { grade: 1 }, name: 'grade_1' }
]
ADS>
```